

The Twelve Apostles

An iconic group of large sea stacks, formed through the erosion of 20-million-year old limestone cliffs



The Twelve Apostles are located in Port Campbell National Park, a coastal area of Victoria. The sea stacks are strung out along about 3 mile (5km) of coastline, with some reaching up to 165 ft (50 m) in height. The Twelve Apostles are one of Australia's best-known geological landmarks, but their name is something of a misnomer, as when they were originally named only nine large stacks existed and one has since collapsed, leaving only eight. However, several prominent headlands and islands nearby are continuing to erode through wave action (see panel, right), so new sea stacks are expected to form in the future. Like all coasts, the shoreline around the stacks is constantly changing.

Until the **1920s**, the stacks were known as the **Sow and Piglets**

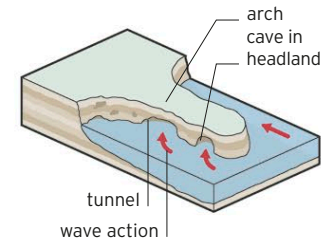
► NEW STACKS IN THE MAKING

This aerial view shows one existing isolated stack, at left, and other headlands and islands that, as they erode, are expected to form new stacks.

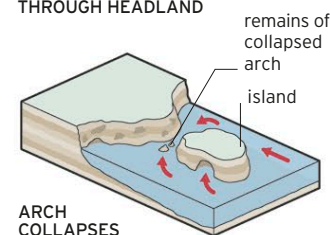


HOW SEA STACKS FORM

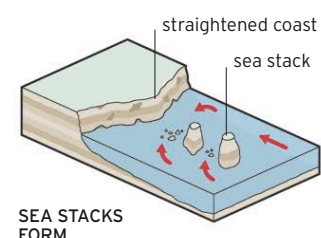
Wave action creates caves in the sides of a headland, eventually tunneling through it. The arch over the tunnel collapses to leave an island, which is then eroded to form stacks.



TUNNEL FORMS THROUGH HEADLAND



ARCH COLLAPSES

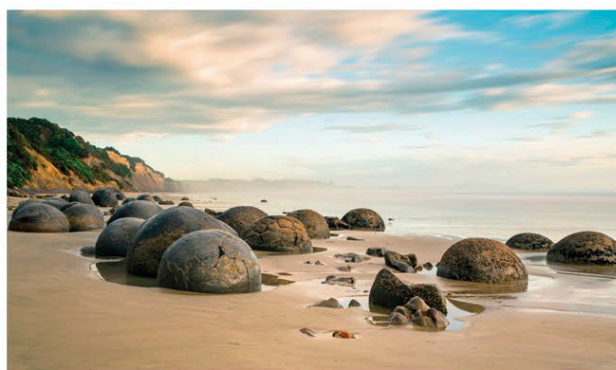


Moeraki Beach

A beach in southeastern New Zealand strewn with some unusually large, almost spherical boulders



Some 45 miles (70km) northeast of Dunedin, Moeraki Beach appears at first sight as though it might have been the location for a giants' game of bowls or petanque. Almost perfectly spherical gray boulders lie scattered across a 165-ft (50-m) stretch of the beach. The scientific view is that the huge boulders actually formed within clay and silt sediments on the seafloor around 60 million years ago by a process called concretion, in which a cementing mineral binds together other mineral particles into a hard, erosion-resistant mass. This process can take millions of years. Later, the concretions, now embedded within mudstone rock, were lifted up, and today most of them are buried within a cliff at the back of Moeraki Beach. There the boulders are gradually exposed and released onto the beach by erosion of the mudstone (see panel, right).

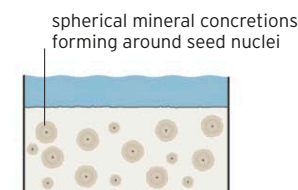


△ SUPERSIZED BOULDERS

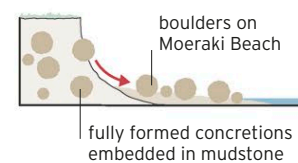
The boulders come in two main sizes—around 3 ft (0.9 m) and 6 ft (1.8 m) in diameter—and weigh several tons. Many are half-buried in the sand.

HOW THE BOULDERS FORMED

The boulders originally grew within marine sediments, which hardened into rock. This was later lifted above sea level and now forms a coastal cliff. From time to time, a boulder is freed up and rolls onto the beach.



FORMATION UNDER THE SEA



TODAY